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Calculation

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In the previous section you formulated a new problem:

Find a value for k such that

$$\begin{cases} \frac{dX}{dt} = kX(1 - X), \\ X(0) = 0.02, \\ X(20) = 0.75 \end{cases}$$

To solve this problem, you are going to use Euler's Method. Make a new copy of Basic_Euler_program.py and adapt it for this problem. First, we have to find an appropriate stepsize.

Exercise A

1/1 point (graded)

To solve the problem, we must select a timestep Δt which for every possible choice of k will lead to valid results if you use Euler's Method.

For now, consider $k = 3$ and $\Delta t = 1$.

What is the value of $X(20)$ using these values?

Use 5 decimals in your answer.

✓ Answer: 1.31251

You have used 1 of 3 attempts

i Answers are displayed within the problem

Exercise B

1/1 point (graded)

Obviously, this value for Δt will not suffice.

Again, consider $k = 3$.

For which values of Δt below are the solutions calculated with Euler's Method plausible? In this case: for which of the following values of Δt do the solutions just increase in time?

Multiple answers may apply.

☐ $\Delta t = 1$

☐ $\Delta t = \frac{1}{2}$

☒ $\Delta t = \frac{1}{\pi}$ ✓

☒ $\Delta t = \frac{1}{4}$ ✓

☒ $\Delta t = \frac{1}{8}$ ✓



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You have used 1 of 2 attempts

i Answers are displayed within the problem

Exercise C

1/1 point (graded)

Now you know which values of Δt are too large, it is time to consider the accuracy. Assume that we want to know the values of X with an error smaller than **0.01**.

For $k = 3$ the steepest part of the graph is near $t = 1.5$ days. So, if the results are accurate enough near $t = 1.5$, they are accurate enough for all other times as well.

Calculate $X(1.5)$ with stepsizes $\Delta t = 0.1, \frac{0.1}{2}, \frac{0.1}{4}, \frac{0.1}{8}, \frac{0.1}{16}$. Write down the results with at least 4 decimals. The error of a result can be estimated as the difference between that result and the result with the doubled stepsize.

What is the largest stepsize for which the estimated error is smaller than **0.01**?

☐ $\Delta t = \frac{0.1}{2}$

☐ $\Delta t = \frac{0.1}{4}$

☐ $\Delta t = \frac{0.1}{8}$

☒ $\Delta t = \frac{0.1}{16}$ ✓

Answer

Correct: $X(1.5) \approx 0.64225$ with an error estimate of $0.64225 - 0.63691 = 0.00534 < 0.01$.

You have used 1 of 3 attempts

i Answers are displayed within the problem

Exercise D

1/1 point (graded)

Using the value for Δt you found above, find an appropriate value for k such that $X(20) = 0.75$. Experiment with values for k . Look for the 2-decimal value for k that gives a value for $X(20)$ that is as close as possible to 0.75 .

The 2-decimal value for k such that $X(20)$ is as near as possible to 0.75 is:

Use 2 decimals in your answer for k .

✓ Answer: 0.25

You have used 1 of 3 attempts

i Answers are displayed within the problem

Now that you have found a value for k , it is time to formulate the results on the next page.

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